TANMOY ROY *PhD*

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SUMMARY

Senior Data Scientist with a PhD and more than 19 years in industry and academia. Builds and deploys on-prem ML models for healthcare at DeliverHealth. The current work includes medical speech-to-text at 12% medical WER and a medical coding co-pilot. On the coding task, a small on-prem model did better than a generic LLM (Gemini). Its AUROC for right and wrong answers increased from about 0.60 to about 0.90.

Started at SIEMENS and worked there for more than 8 years with C, C++, Java, and Oracle. Then completed a PhD in three years and a post-doctoral fellowship at the University of Johannesburg. The research topic was speech emotion recognition. Wrote eight peer-reviewed publications.

CORE STRENGTHS

- **Production ML.** Built Med-ASR, a medical coding co-pilot, a RAG correction engine, and physician note generation. Earlier, built a claim denial model (XGBoost, 89% accuracy) and an X-ray disease detector (ResNet34, 84%).
- **Model evaluation.** Used white-box logits as a confidence meter. Compared a fine-tuned SLM with a Gemini baseline. Compared an attention-based text classifier with Naive Bayes, logistic regression, random forest, and MLP. Teaches hypothesis tests, feature selection, and outlier detection.
- **ML engineering.** Ran inference on on-prem GPUs. Fine-tuned Gemini-1.5-Pro and Whisper on proprietary data, and the audio did not go to a public cloud. Tools are unsloth, Docker, KubeFlow, AWS, and GCP. Also teaches MLOps: packaging, CI/CD, and model serving with Flask.
- **Research.** Completed a PhD in speech emotion recognition in three years on a Global Excellence Stature (GES) scholarship. Proposed the SEGRT feature set and a speech endpoint-detection algorithm. Guides a PhD scholar from King Saud University and an MTech student from BITS Pilani. Reviews for the *International Journal of Systems Science*.
- **Software engineering.** For more than 8 years at SIEMENS, designed and implemented telecom provisioning, billing, and charging modules. Used C/C++, ProC, Java Spring, and Oracle PL/SQL on Linux.
- **Retrieval (RAG).** Built a RAG engine that corrects ASR drafts before delivery. This decreases manual cleanup. Tools are ollama, langchain, and langgraph. Also used record linkage to remove duplicate patient records.
- **Data science and finance.** Led a team of data scientists, quantitative analysts, and software engineers. Built a production backtest for complex option strategies, and SHAP and ALE plots for model transparency. Predicted 30-day readmission from EHR data with 88% test accuracy.

EXPERIENCE

DeliverHealth Solutions

Remote, India [July '24 – Present]

Senior Data Scientist

Leads AI/ML product ideas and their adoption across the company. Designs and deploys on-prem specialist models with product owners and with partners such as Google and Microsoft. The goal is small models that a person can inspect and that can defer a decision. Fluency alone is not the goal.

- **Medical coding co-pilot.** Fine-tuned an SLM on proprietary historical data in a continuous learning loop. The model helps human coders and decreases the time for each document. White-box logits give a confidence meter. On this task, the on-prem SLM did better than a generic LLM (Gemini). The AUROC for right and wrong answers increased from about 0.60 to about 0.90.

- **Medical ASR (Med-ASR).** Built an in-house speech-to-text engine on proprietary data. Fine-tuned Gemini-1.5-Pro and Whisper. The engine reached 12% medical WER, and the encounter audio did not go to a public cloud.
- **RAG correction engine.** Built a retrieval-augmented engine that corrects ASR drafts before delivery. It decreases manual cleanup and increases document quality.
- **Medical note generation.** Built a template-based tool that writes physician notes. It runs on a self-service platform and uses Gemini 3.1 Pro and proprietary historical data.

Cloudcraftz Solutions

Kolkata, India [Nov '22 – Jun '24]

Senior Data Scientist

Led and mentored a team of data scientists, quantitative analysts, and software engineers. The team delivered systems for the finance domain.

- **Option strategy backtest.** Built a production backtest for complex option strategies. The work combined software engineering, financial analytics, and data science.
- **Explainable AI.** Built the full product. It shows SHAP and accumulated local effect (ALE) plots for model transparency.
- **Data visualization platform.** Led the development of an analytics UI that needs little manual work.

Veersa Technologies

Kolkata, India [May '22 – Oct '22]

Data Scientist (Consulting)

Built ML models on tabular and image data for healthcare projects. Also did data engineering, put ML models into production, and helped build the ML team.

- **Claim denial prediction.** Trained an XGBoost classifier to predict if a health institute's claims get denied or approved. Accuracy was 89%. The work also showed the reasons for denials.
- **X-ray disease detection.** Used a ResNet34 model. Accuracy was 84%.
- **Patient record de-duplication.** Used record linkage to join broken chains of patient information.
- **Clinical trial enrollment.** Built a proof of concept on public data that predicts final enrollment. Accuracy was about 75%.
- **Handwritten medical forms.** Built a form reader around AWS Textract output.

Research and Training Consultancy

Kolkata, India [Feb '20 – Present]

Research Consultant

Research work in computer vision, NLP, and data science.

- **Human fall detection.** Evaluated current deep learning modules. Proposed a Masked R-CNN segmentation model [2] with 94% F1 and 89% IoU.
- **Text classification.** Built a multi-head attention classifier [1] with more than 96% test accuracy. Compared it with Naive Bayes, logistic regression, random forest, and MLP.
- **EHR readmission.** Predicted 30-day readmission from EHR data with 88% test accuracy. Used process-mining features in sequence-to-sequence models.
- **Medical imaging and CV.** Used U-Net to detect polyps in colonoscopy images. Used MobileNet V2 and Inception ResNetV2 to detect face masks in surveillance video.

Training Consultant

Teaches students, people who change careers, and engineers in industry. The topics are data science, machine learning, and deep learning (ANN, CNN, RNN, attention). Other topics are Python for ML (numpy, pandas, PyTorch, TensorFlow, Flask) and MLOps. Cloud topics are Docker and AWS (DMS, RDS, EMR, SageMaker).

University of Johannesburg

Johannesburg, South Africa [Jul '16 – Sep '21]

Post Doctoral Research Fellow [Oct '19 – Sep '21]

Built a feed-forward network [3] on the SER feature set from the PhD [8]. Wrote two surveys. One is on emotion recognition in the fourth industrial revolution [5]. The other is on SER classification techniques [4]. Started research work with King Saud University [2] and BITS Pilani [1].

Doctoral Research Fellow [Jul '16 – Jun '19]

Thesis: "Detecting emotions from speech using Machine Learning Techniques." Completed the PhD in three years. Proposed SEGRT, the *subjective emotional gap reduction technique* [8, 6]. Also proposed a speech endpoint-detection algorithm for pre-processing [7].

SIEMENS

Kolkata, India [Jan '07 – Sep '15]

Associate Consultant [Jan '13 – Sep '15]

• Designed and implemented a Telecom Service Provisioning system with *Java Spring* and *Oracle*. It was a multi-tier application on RedHat Linux. Later customized it with *Java, SOAP*, and *XML*.
• Designed and implemented two modules for a GSM-R billing system. The CRM integration module used *Java* and *Oracle PL/SQL*. The billing module used *C/C++* and *Oracle PL/SQL* on *Linux*.

Sr. System Engineer [Jan '11 – Dec '12]

• Designed and implemented a charging module for an electronic car billing system in *C/C++* and *Oracle*.
• Built a GUI for telecom call analysis with *Qt* and *C++*. It tracked calls and showed zone-based analysis with graphical simulations.

System Engineer [Jan '09 – Dec '10]

• Designed and implemented Bill Printing and Bill By Email modules for a telecom billing system. Used *Java, ProC/C++*, and *Oracle PL/SQL*.
• Designed and implemented a charging module for telecom interconnect billing with *Java* and *Oracle*.

Associate System Engineer [Jan '07 – Dec '08]

• Gave system support alone for an IVR and USSD solution. Used *Java, C++, Linux*, shell, and awk.
• Built an automated monitoring system with *Java*, shell, and awk to improve the client experience.

TECHNICAL EXPERTISE

- **Programming Languages:** Python, C/C++, R, Matlab, Java, JavaScript, SQL, Scala, Julia.
- **Libraries:** ollama, langchain, langgraph, unsloth, pytorch, tensorflow, pandas, matplotlib, scipy, opencv, pyspark.
- **Technologies/Platforms:** On-prem GPU inference, Cloud (AWS, GCP), Linux, Docker, MLOps (KubeFlow), REST, SOAP, XML.
- **Databases:** RDBMS (Oracle, PostgreSQL, SQL Server, MySQL), NoSQL (MongoDB, DynamoDB, Cassandra), Cloud DB (Amazon RDS).

EDUCATION

Doctor of Philosophy – University of Johannesburg, 2019. Applied ML & Speech Signal Processing.
Master of Science – Lovely Professional University, 2013. Information Technology.
PG Diploma in Computer Applications – DOEACC Society, 2004.
Bachelor of Science – University of Calcutta, 2002. Economics (Hons), CS, Mathematics.

SCHOLARSHIPS, REVIEWING AND CERTIFICATES

- Global Excellence Stature (GES) Doctoral Fellowship, University of Johannesburg, 2016.
- GES Post Doctoral Fellowship, University of Johannesburg, 2019.
- Journal Reviewer, *International Journal of Systems Science* (Taylor and Francis).
- Coursera: Deep Learning with PyTorch: Neural Style Transfer; Neural Network from Scratch in TensorFlow.

PUBLICATIONS

[1] Sinha J and Roy T. "Attention Based Encoder Architecture for Automatic Text Classification: A Case Study on Text-news Classification". In: 2022. DOI: DOI:10.55041/IJSREM12107.

[2] Alanazi T and Roy T. "Generating Human Segmentation Dynamically in Multi-camera Setup Using Deep Learning". In: *Draft*. 2022.

[3] Roy T, Marwala T, and Chakraverty S. "Chapter 12 - Speech emotion recognition using deep learning". In: *New Paradigms in Computational Modeling and Its Applications*. Ed. by Snehashish Chakraverty. Academic Press, 2021, pp. 177–187. ISBN: 978-0-12-822133-4. DOI: https://doi.org/10.1016/B978-0-12-822133-4.00009-8.

[4] Roy T, Marwala T, and Chakraverty S. "A Survey of Classification Techniques in Speech Emotion Recognition". In: *Mathematical Methods in Interdisciplinary Sciences*. John Wiley & Sons, Ltd, 2020. Chap. 3, pp. 33–48. ISBN: 9781119585640. DOI: https://doi.org/10.1002/9781119585640.ch3. eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/9781119585640.ch3.

[5] Roy T, Marwala T, and Chakraverty S. "Advancements and Role of Emotion Recognition in the 4th Industrial Revolution". In: *The Disruptive Fourth Industrial Revolution: Technology, Society and Beyond*. Ed. by Wesley Doorsamy, Babu Sena Paul, and Tshilidzi Marwala. Cham: Springer International Publishing, 2020, pp. 179–203. ISBN: 978-3-030-48230-5. DOI: 10.1007/978-3-030-48230-5_8.

[6] Roy T, Marwala T, and Chakraverty S. "Speech Emotion Recognition Using Neural Network and Wavelet Features". In: *Recent Trends in Wave Mechanics and Vibrations*. Ed. by S Chakraverty and Paritosh Biswas. Singapore: Springer Singapore, 2020, pp. 427–438. ISBN: 978-981-15-0287-3. DOI: 10.1007/978-981-15-0287-3_30.

[7] Roy T, Chakraverty S, and Marwala T. "Precise detection of speech endpoints dynamically: A wavelet convolution based approach". In: *Communications in Nonlinear Science and Numerical Simulation* 67 (2019), pp. 162–175. ISSN: 1007-5704. DOI: https://doi.org/10.1016/j.cnsns.2018.07.008.

[8] Roy T et al. "Introducing New Feature Set based on Wavelets for Speech Emotion Classification". In: *2018 IEEE Applied Signal Processing Conference (ASPCON)*. 2018, pp. 124–128. DOI: 10.1109/ASPCON.2018.8748666.